Ontology for

CAREER IN AI

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CM4650 - Semantic Web & Ontological Modelling
Department of Computational Mathematics
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PROBLEM

- Very difficult to start AI career with all the misinformation. Lacks a proper roadmap.

- What you should learn to learn another subject?
- What career options you have?
- What technologies to learn?
- What would be the pay looks like?

With an **ontology** you can make “connections” and answer them

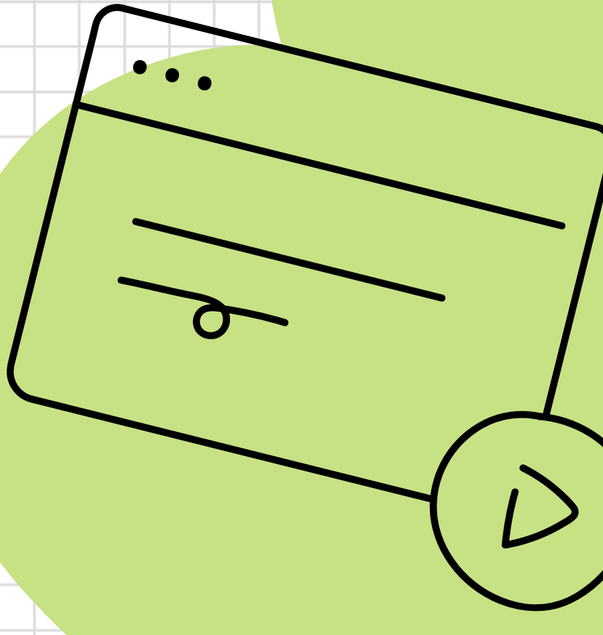
ONTOLOGY DESIGN SOURCES

ACM/IEEE CS-AAAI curricula 2023

UOM BSc (Hons) AI curriculum

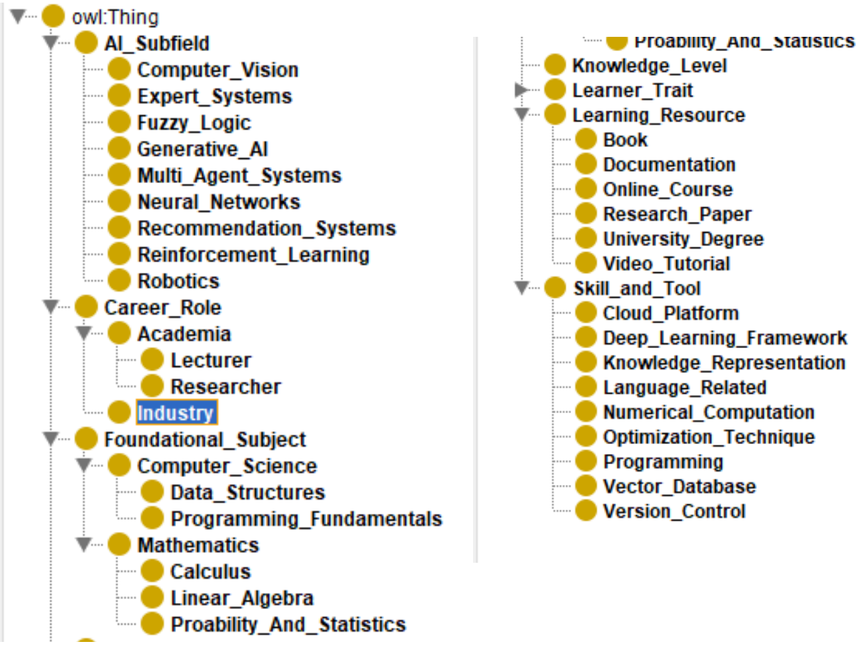
Coursera, udemy courses

Industry, job market



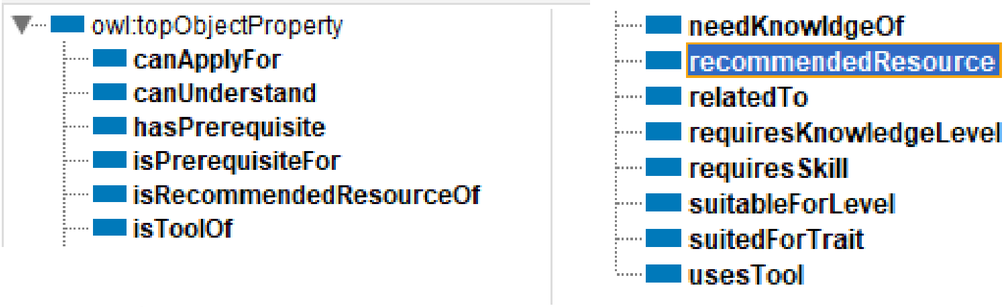
DESIGN

CLASSES



- 45 classes
- 208 individuals (instances)

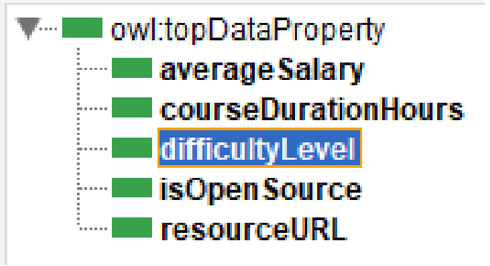
OBJECT PROPERTIES



- 14 object properties

DATA PROPERTIES

- 5 data properties



INDIVIDUALS

- academic_advisor
- actor_critic
- adam_optimizer
- advanced
- advanced_computer_vision
- advanced_cv
- agentic_ai
- ai_foundations_course
- ai_researcher
- ai_solution_architect
- algorithms
- arrays_and_lists
- attention_is_all_you_need
- autonomous_navigation
- aws_codecommit
- aws_sagemaker
- azure_machine_learning
- bayesian_optimization
- bayesian_statistics
- beginner
- bsc_hons_ai
- buidling_agentic_ai_systems
- c
- calculus_I
- calculus_II
- cloud_architect
- collaborative_filtering
- competitive_agents
- complex_system_modelling
- complexity
- computer_networks
- computer_vision_advanced
- computer_vision_playlist
- computer_vision_youtube
- content_based
- control_flow
- convolution_neural_networks
- cooperative_agents
- critical_reasoner
- csharp
- curiosity_driven
- cv_researcher
- cv_specialist
- data_architect
- data_driven
- data_scientist
- data_scientist_senior
- data_scientist_senior
- data_types
- database_systems
- deep_learning_specialization_course
- deep_neural_networks
- department_chair
- description_logic
- differential_calculus
- discrete_math
- distributed_planning
- dsa_I
- dsa_II
- eigenvalues
- evidence_seeker
- experimental_mindset
- expert_system_shells
- expert_systems_principles
- faiss
- feedforward_networks
- functions_and_modules
- fuzzy_control
- fuzzy_inference
- fuzzy_intermediate_course
- fuzzy_logic_base
- fuzzy_logic_int
- fuzzy_sets_basics
- genai_specialist
- genetic_algorithms
- git
- github
- gitlab
- google_vertex_ai
- grid_search
- hybrid_recommenders
- hypothesis_driven
- hypothesis_testing
- ibm_watson_studio
- image_classification
- image_classification_enhanced
- industrial_robotics
- inference_engines
- infrastructure_architect
- integral_calculus
- intermediate
- intro_ai_playlist
- introduction_to_fuzzy_logic
- java

http://www.s

CONSTRAINTS

01
Domain & Range

Description: recommendedResourc

Equivalent To

SubProperty Of

Inverse Of

isRecommendedResourceOf

Domains (intersection)

AI_Subfield

Ranges (intersection)

Learning_Resource

02
Characteristics

Characteristics: recomm

- [ ] Functional
- [ ] Inverse functional
- [ ] Transitive
- [ ] Symmetric
- [x] Asymmetric
- [ ] Reflexive
- [x] Irreflexive

03
Cardinality
Existential
Universal
Exactly

Description: AI_Subfield

Equivalent To

SubClass Of

hasPrerequisite some Foundational_Subject

usesTool min 2 Skill_and_Tool

04
Disjointness

Description: Industry

Equivalent To

Disjoint With

Academia

COMPETENCY QUESTIONS

CQ1 What knowledge does a specific role require?

CQ2 Which resources are recommended for a topic?

CQ3 What foundational subjects are needed for what?

CQ4 Which frameworks are used in which subfields?

CQ5 What are high-paying roles and their skills?

CQ6 What skills are missing between roles?

CQ7 How many subfields per knowledge level?

CQ8 What topics relate to a specific field?

CQ9 What learning materials exist for what topics?

CQ10 Complete 360-degree career overview

DEVELOPMENT

TECHNOLOGY

- Protege → develop
- GraphDB → host
- Streamlit → frontend
- SPARQL Wrapper
- Ollama

FRONT-END

1. Run competency questions → Ensure ontology works as supposed to
2. User area → User dashboard to cover user requirements → queries are hidden
3. Chatbot → KG-RAG chatbot using local models. Chat with the graph
4. Playground → Run your own query

DEMO